

Space Development Agency
Tranche 2 Transport Layer Proliferated Warfighter Space Architecture
Payload Processing Facility Support

Request for Information

The Space Development Agency (SDA) seeks U.S. corporation industry feedback on commercial Payload Processing Facility (PPF) support for SDA spacecraft, satellites, or payloads. The Contractor will provide PPF support to include advance planning, facilities, supplies, commodities, and services necessary for the SV customers and Launch Service Providers (LSPs) to perform prelaunch processing and integration of the SV and LSP hardware. The Contractor shall designate a facility located on or near Kennedy Space Center (KSC)/Cape Canaveral Space Force Station (CCSFS) or Vandenberg Space Force Base (VSFB), or both.

Responses to this RFI will specifically inform SDA's Transport Layer Tranche 2 Launches:

Mission	Dates	Location
T2TL-A	Aug/Sept 2026	VSFB
T2TL-B	Sept/Oct 2026	VSFB
T2TL-C	Oct/Nov	CCSFS
T2TL-D	Oct/Nov	CCSFS
T2TL-E	Dec 2026/Jan 2027	VSFB
T2TL-F	Jan/Feb 2027	VSFB
T2TL-G	Feb/Mar 2027	VSFB
T2TL-H	Mar/Apr 2027	VSFB
T2TL-I	Apr/May 2027	CCSFS
T2TL-J	May/Jun 2027	CCSFS

Overview

The PWSA's operational utility is predicated on the availability of a ubiquitous data and communications Transport Layer provided by a proliferated constellation of relatively small, mass-producible space vehicles (SVs) in low Earth orbit (LEO). The Tranche 2 Transport Layer (T2TL) SVs will be similar to those currently under development for the Tranche 1 Transport Layer (T1TL) and Tranche 1 Development and Experimentation System (T1DES), with targeted technology enhancements, mission-focused payload configurations, increased integration, and greater production efficiencies. T2TL will provide global communications access and deliver persistent regional encrypted connectivity in support of Warfighter missions around the globe. Specifically, T2TL features multiple SV and mission configuration variants procured through a multi-solicitation and multi-vendor acquisition approach. The T2TL establishes initial launch capability (ILC) in September 2026 with the launch of a plane of T2TL-Beta SVs and continues with an approximately year-long, monthly launch campaign involving the multiple T2TL variants. As these SVs are placed into their insertion orbits, SDA will undergo a continuous checkout and commissioning process to prepare Tranche 2 for acceptance into operations conducted out of the Operations Centers (OCs) at Grand Forks, AFB (OC-N) and Redstone Arsenal, AL (OC-S). The Transport Layer is an element of the PWSA with four major tenets: low latency/high volume data transfer, sensor-to-shooter support, TACSATCOM applications, and the JADC2 backbone.

SDA's Request for Information

SDA is explicitly seeking the following information to inform future solicitations and reduce risk toward PPF support to include advance planning, facilities, supplies, commodities, and services necessary for the SV customers and Launch Service Providers (LSPs) to perform prelaunch processing and integration of the SV and LSP hardware. SDA strongly encourages industry to provide feedback to the following specific questions to inform Tranche 2 Transport Layer future launches.

Payload Processing Facility

The Contractor shall perform advance planning necessary to furnish facilities, supplies, commodities, and services necessary for processing the SV(s) and providing for launch vehicle integration operations, when required. The Contractor must be able to provide the following primary capabilities: (1) facilities operated and maintained in accordance with safety, cleanliness, and security standards to protect the assets processed therein; (2) facilities and support to satisfy mission-specific requirements; (3) planning and documentation to optimize the SV use of its facility; (4) commodities and services necessary to support SV processing; and, (5) facilities and services to house support personnel.

Please address the following:

1. Communication: Provide, operate, and maintain the necessary equipment to handle the flow of voice, video, timing, telemetry and networks, both internal and external to the processing facility, during SV processing and launch.
2. Fueling and Servicing Available: For payloads that require propellants or other consumables, the PPF provides a safe and controlled environment for fueling and servicing.
3. Encapsulation space: The payload is enclosed within a fairing or protective shell that shields it during ascent through the atmosphere. This process takes place in the PPF.
4. Cleanrooms: To prevent contamination of sensitive instruments, PPFs often incorporate cleanrooms with strict controls on particulate matter and humidity. For areas with seismic activity, the Contractor shall make provisions for the SV to bolt down the support stand and associated hardware (if required). The SV cleanroom areas shall have static dissipative flooring.
5. Environmental Controls: Temperature and humidity are carefully controlled to protect the payload.
 - a. Cleanliness: Stringent cleanliness standards are crucial, often requiring Class 100 to Class 10,000 cleanrooms (or even better for some payloads) to minimize contamination by dust, particles, and other foreign objects that could damage sensitive instruments.
 - b. Temperature and Humidity Control: Precise temperature and humidity control are essential to maintain the integrity of payload components, especially sensitive electronics and optical instruments.
 - c. Electrostatic Discharge (ESD) Protection: ESD can damage electronic components, so the PPF must have measures in place to prevent electrostatic buildup and discharge.
6. Safety and Security:
 - a. Hazardous Materials Handling: Many payloads involve hazardous materials like propellants and pressurized gases. The PPF must have appropriate safety systems and procedures for handling, storing, and disposing of these materials.
 - b. Fire Suppression: Robust fire suppression systems are necessary to protect personnel and equipment in the event of a fire.

- c. Security: Access to the PPF must be restricted to authorized personnel to protect sensitive technology and prevent sabotage.
- 7. Infrastructure and Equipment:
 - a. Lifting and Handling Equipment: Heavy-duty cranes, hoists, and other lifting equipment are required to move and manipulate large and heavy payloads.
 - b. Power and Data Connections: Reliable power and data connections are essential for testing and operating the payload. Electrical systems are designed, operated, and maintained in accordance with the National Electrical Code (NEC) and the National Fire Protection Association (NFPA). For facilities capable of performing propellant loading, the Contractor shall provide per NEC Class I, Division 2, Group C and D hazard proofing for both the cleanroom and airlock electrical power receptacles.
 - c. Cleanrooms and Clean Support Areas: Cleanrooms are required for assembly, integration, and testing. Adjacent clean support areas are needed for cleaning and preparing equipment and materials before they enter the cleanroom.
 - d. Transportation Interface: The PPF must have a suitable interface for transporting the processed payload to the launch pad, often involving specialized transporters.
- 8. Provide a contract type preference, i.e., IDIQ, BPA, other.

In summary, a payload processing facility plays a crucial role in preparing spacecraft payloads for launch, ensuring their safety and functionality in the challenging environment of space.

SUBMISSION INSTRUCTIONS

- i. All responses to this RFI must be received no later than 1700 EST 3 May 2025.
- ii. Responses should be no more than 10 pages (exclusive of lists of tables, figures, acronyms, etc.) and should use a minimum of 12-point font, single line spacing and 1-inch margins.
- iii. Responses shall be sent electronically to ussf.pentagon.sda.mbx.sn-25-0016@mail.mil.
- iv. Responses to this notice shall include Vendor's Company Name, Address, and Contact Information.
- v. When submitting a response, please use the subject PPF Services RFI [-Vendor Name]".
- vi. Any responses submitted through different channels or to a different email address will not be considered.
- vii. Organizations under a common corporate banner, even if under separate CAGE Code, are encouraged to limit to a single response.

viii. DISCLAIMERS AND IMPORTANT NOTES

This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed. This is an RFI issued solely for information and planning purposes; it does not constitute a formal solicitation for proposals. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the government to form a binding contract. Submission of a response by U.S. corporations is strictly voluntary and is not required to propose to a subsequent SDA solicitation. No solicitation exists; therefore, do not request a copy of the solicitation. If a solicitation is released, it will be synopsized on the SAM.gov website. It is the responsibility of any potential offerors/bidders to monitor this site for release of any solicitation or synopsis. The SDA will NOT provide reimbursement for costs incurred in responding to this RFI or participating in any subsequent workshop. SDA will acknowledge receipt of all submissions but is under no obligation to provide feedback.

To the maximum extent possible, please submit non-proprietary information. If proprietary information is submitted, it must be appropriately and specifically marked. It is the submitter's responsibility to clearly

define to the Government what is considered proprietary data. Any proprietary information should be clearly labeled as "Proprietary." The SDA will not publicly disclose proprietary information obtained as a result of the RFI. To the full extent that it is protected pursuant to the Freedom of Information Act and other laws and regulations, information properly identified by a respondent as "Proprietary" will be appropriately controlled. Submissions may be reviewed by Government personnel and support contractors bound by appropriate non-disclosure agreements. Responses to this RFI will not be returned. Intellectual or other privileged or proprietary information contained in responses to this RFI will not be distributed outside of the Department of Defense or United States Government employees from other Government agencies who are working with the SDA on this RFI.